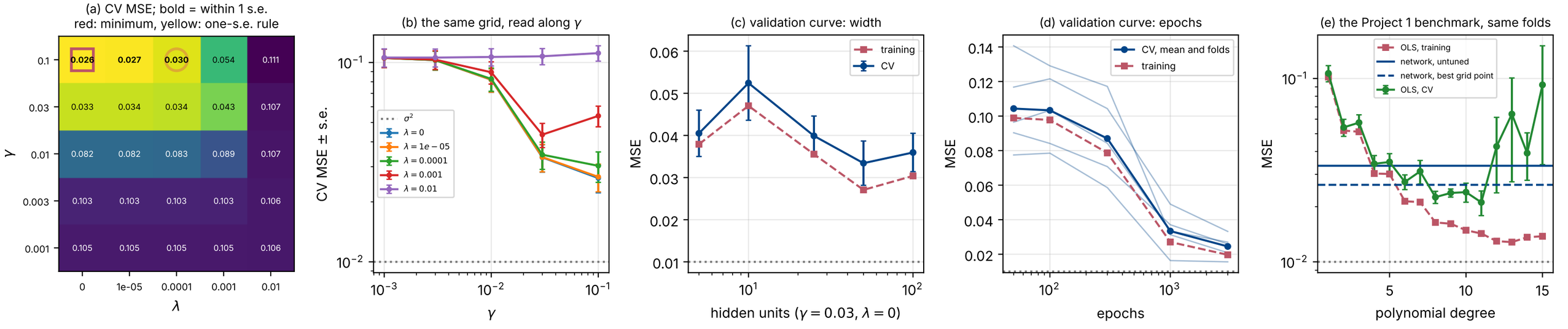
(a) CV MSE; bold = within 1 s.e.
red: minimum, yellow: one-s.e. rule
γ
λ
0.1
0.03
0.01
0.003
0.001
0
1e-05
0.0001
0.001
0.01
0.026
0.027
0.030
0.054
0.111
0.033
0.034
0.034
0.043
0.107
0.082
0.082
0.083
0.089
0.107
0.103
0.103
0.103
0.103
0.106
0.105
0.105
0.105
0.105
0.106
(b) the same grid, read along γ
CV MSE ± s.e.
γ
σ²
λ = 0
λ = 1e − 05
λ = 0.0001
λ = 0.001
λ = 0.01
(c) validation curve: width
MSE
hidden units (γ = 0.03, λ = 0)
training
CV
(d) validation curve: epochs
MSE
epochs
CV, mean and folds
training
(e) the Project 1 benchmark, same folds
MSE
polynomial degree
OLS, training
network, untuned
network, best grid point
OLS, CV